Office hours

Carver 379

Today: Fix a nonempty set V. Let $\mathcal{T}_V$ denote the set of all trees on vertex-set V.

Theorem (Cayley's formula). *If* $|V| = n \geqslant 2$, *then* $|\mathcal{T}_V| = n^{n-2}$.

Note: We are counting all trees on V, not trees up to isomorphism.

The strategy is the following:

Step 0: consider $V^{n-2} = \{(x_1, x_2, \ldots, x_{n-2}) : x_i \in V\}$, all $(n-2)$-tuples in V, which has $|V^{n-2}| = |V|^{n-2} = n^{n-2}$.

Step 1: construct a function $\text{Prüfer}_V : \mathcal{T}_V \longrightarrow V^{n-2}$.

Step 2: show that Prüfer_V is a bijection from $\mathcal{T}_V$ to V^{n-2}. This will imply Cayley's formula.

To define Prüfer_V, fix an ordering on $V = \{v_1, v_2, \ldots, v_n\}$ so that $v_1 < v_2 < \cdots < v_n$. $V = \{1, 2, \ldots, n\}$

Definition (Prüfer codes). *For an ordered set* V *with* $|V| = n \geqslant 2$, *we define a function*

$$\textit{Prüfer}_V : \mathcal{T}_V \longrightarrow V^{n-2}$$

recursively on n *as follows:*

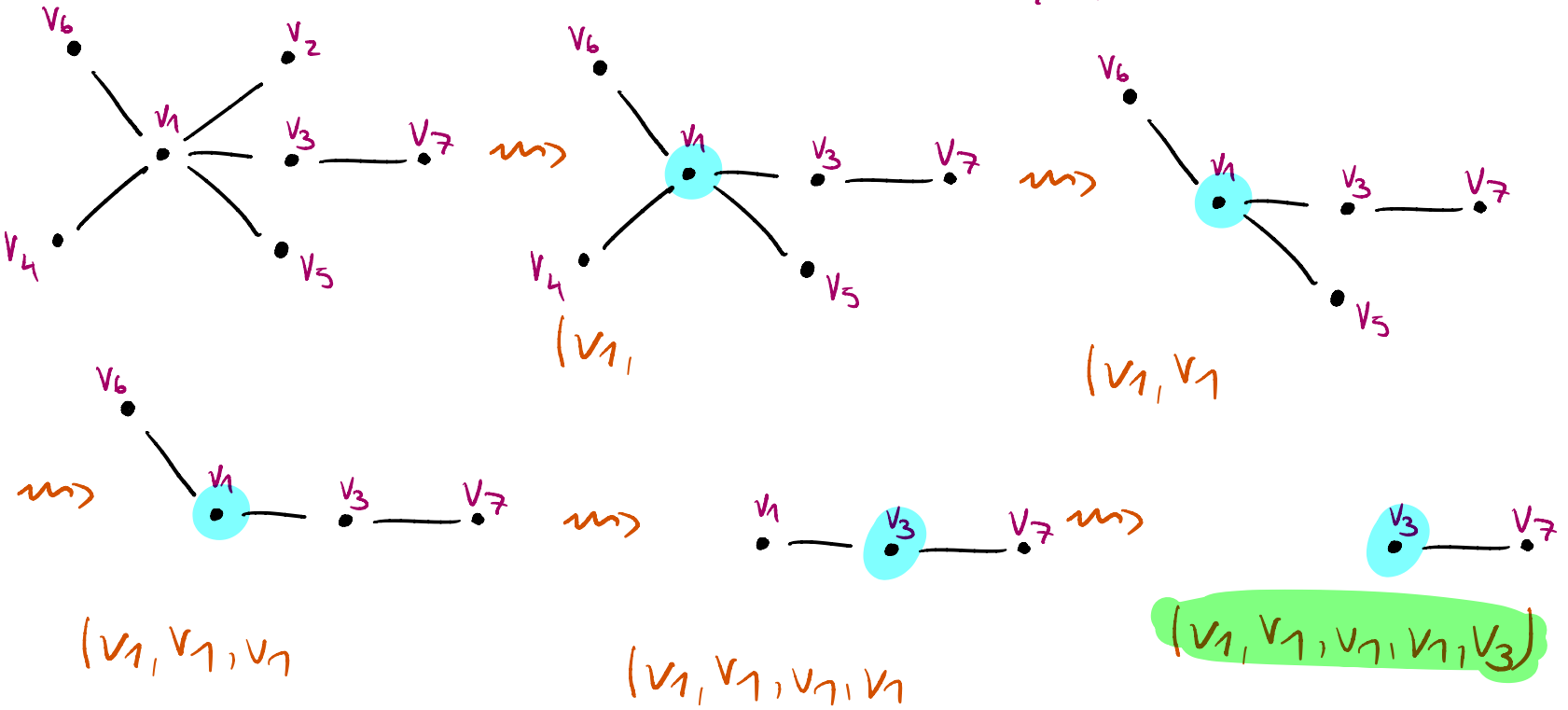
(1) *If* $n = 2$ *define* $\textit{Prüfer}_V(T) = ()$, *the empty tuple;* $T = K_2$!

(2) *If* $n \geqslant 2$, *given* $T \in \mathcal{T}_V$ *let* $\ell \in V$ *denote the smallest leaf of* T *(with respect to the ordering on* V*). Let* v *denote* ***the*** *neighbor of* ℓ *in* T *and define*

$$\textit{Prüfer}_V(T) = (v, \textit{Prüfer}_{V-\{\ell\}}(T - \ell)).$$

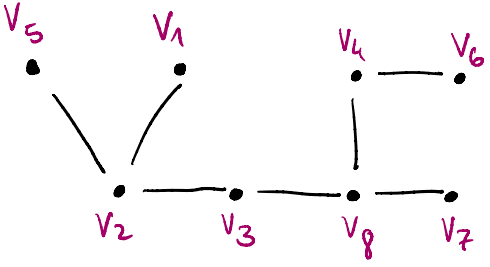
Informally: to build $\text{Prüfer}_V(T)$ delete the smallest leaf, write down its neighbor, and repeat until there are only two vertices left.

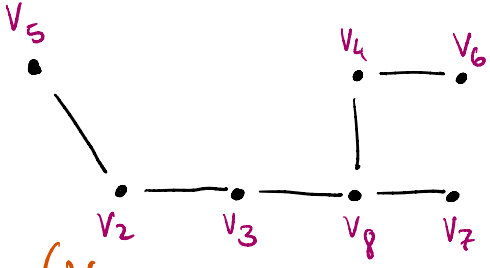
Example. *Compute the Prüfer codes of the following trees.* $V = \{v_1, \dots, v_7\}$, $v_1 < \dots < v_7$

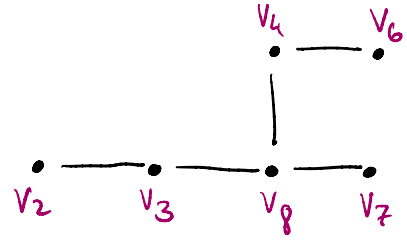


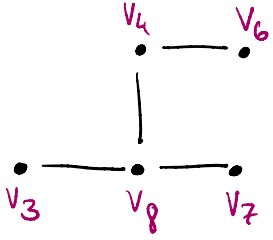
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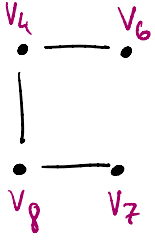
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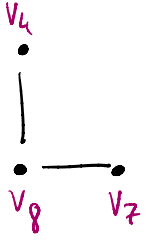














$(v_2, v_2, v_3, v_8, v_4, v_8)$

Lemma. *Fix an ordered set V with $|V| = n \geq 2$. For any $v \in V$ and $T \in \mathcal{T}_V$, the entry v appears exactly $\deg_T v - 1$ times in the sequence $\text{Prüfer}_V(T)$.*

Proof: Induction on n

Base case $n=2$: $T = K_2 =$ 

, $\text{Pr}_{\{v_1,v_2\}}(T) = (\ )$

So v_i appears 0 times and $0 = \deg_T v_i - 1$ ✓

Assume $n \geq 3$. Take V ordered set on n-elements

Let $T \in \mathcal{T}_V$. Let ℓ denote the smallest leaf of T. Let w its neighbor

Let $v \in V$. WTS: v appears $\deg_T v - 1$ in $\text{Pr}_V(T)$

Case 1: $v = \ell$. Then v appears 0 times in $\text{Pr}_V(T)$. Since also $\deg_T v - 1 = 0$, we are done.

Case 2: $v \neq \ell$. Thus $v \in T - \ell$ and by IH v appears $\deg_{T-\ell} v - 1$ times

in $Pr_{T-e}(T-\ell)$

But $Pr_V(T) = (w, Pr_{T-e}(T-\ell))$

↳ v appears $\deg_{T-e} v - 1$ Times

Case 2.1: $v \neq w, \ell$

Thus v appears $\deg_{T-e} v - 1$ Times in $Pr_v(T)$

But $\deg_T v = \deg_{T-e} v$ so we are done

Case 2.2: $v = w$

Thus v appears $1 + \deg_{T-e} v - 1$ Times in $Pr_v(T)$

But $\deg_T v = 1 + \deg_{T-e} v$ so we are done ✓

Theorem. *For any ordered set V with $|V| = n \geqslant 2$, the function Prüfer_V is a bijection from $\mathcal{T}_V$ to V^{n-2}.*

Proof: Induction on n

Base case n=2: Only one Tree (K_2) only one Tuple: () ✓

Assume V has $|V| = n \geqslant 3$

Step 1: Pr is injective.

Let $T, S \in \mathcal{T}_V$ such that $Pr_V(T) = Pr_V(S) = (s_1, \ldots, s_{n-2})$

Let t = smallest leaf of T, s = smallest leaf of S

Claim: $t = s$

By definition, t never appears in $Pr_V(T) = Pr_V S$

Thus by lemma, $\deg_S t = 1$, i.e: t is a leaf for $S \Rightarrow s \leqslant t$

Symmetric: $t \leqslant s$. Hence $t = s$.

Hence $Pr_{V-\tau}(T-\tau) = (s_2, \ldots, s_{n-2}) = Pr_{V-s_1-t}(S-t)$

By IH $T-\tau = S-t$

Since $\tau = s_1$ is attached only to s_1 in T and in S

We have $T = S$.

STEP 2: Pr_V is surjective. Take $(s_1, s_2, \ldots, s_{n-2})$ in V^{n-2}

WTS: $\exists\ T \in \mathcal{T}_V$ such that $Pr_V(T) = (s_1, \ldots, s_{n-2})$.

Let $v \in V$ be the smallest element not in $\{s_1, \ldots, s_{n-2}\}$.

Now $(s_2, s_3, \ldots, s_{n-2}) \in (V - \{v\})^{n-3}$

By IH, There is $T' \in \mathcal{T}_{V-\{v\}}$ such that $Pr_{V-\{v\}} = (s_2, \ldots, s_{n-2})$

Let $T = T' + v + vs_1$

Claim: v is the smallest leaf of T. It is a leaf by definition ✓

Let w denote the smallest leaf of T, and assume $w \neq v$ $(w < v)$

Then w is a leaf in $T-v$. By lemma, w never appears in $Pr_{V,v}(T-v)$

$Pr_{V,v}(T-v) = (s_2, \ldots, s_{n-2})$. Hence $w \notin \{s_2, \ldots, s_{n-2}\}$

And also s_1 is not a leaf in T, because it has a neighbor in T' and also a new one v in T. Thus $s_1 \neq w$.

Hence $w \notin \{s_1, s_2, \ldots, s_{n-2}\}$ with $w < v$

Contradicts definition of v!

Hence v is the smallest leaf of T.

Now $Pr_V(T) = (s_1, Pr_{V,v}(T-v)) = (s_1, s_2, \ldots, s_{n-2})$ ∎

According to the previous theorem, to any given sequence $P \in V^{n-2}$ one can associate a unique $T \in \mathcal{T}_V$.

Algorithm: Initialize with the empty graph $F = (V, \varnothing)$ and $S = \varnothing$, and iterate the following:

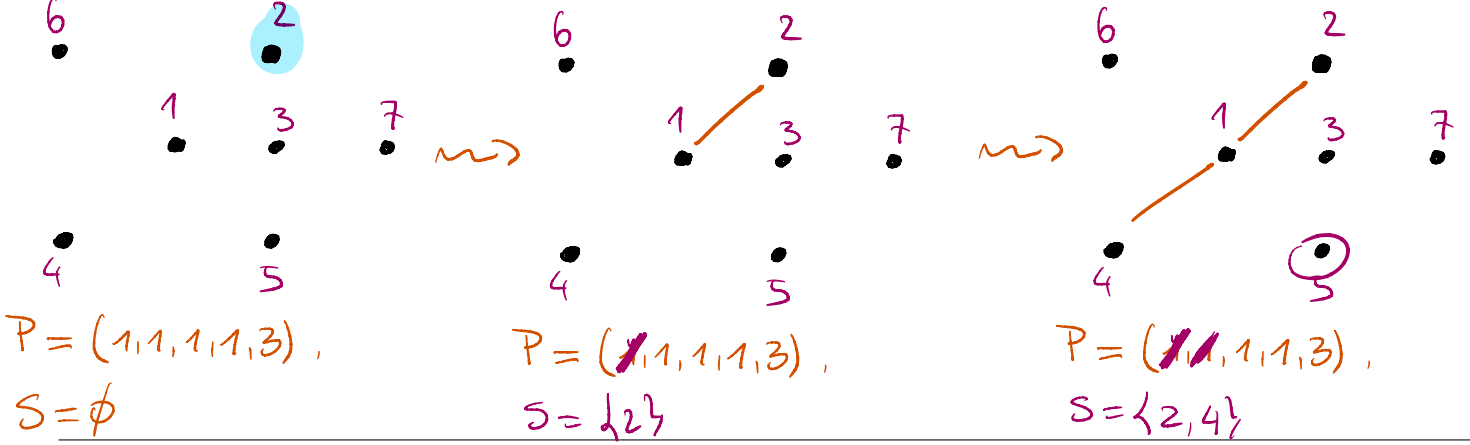
(1) Let $v \in V$ the smallest element which does not appear in S nor in P.

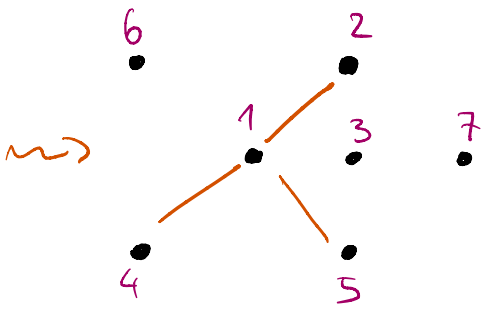
(2) If p denotes the first entry in P, add the edge vp to F.

(3) Remove the first entry of P (so we now have a sequence of length one fewer) and add v to S.

After $n-2$ iterations, P is the empty sequence and $|S| = n-2$. Thus $S = V - \{u, v\}$. Adding the edge uv to F yields the desired sequence.

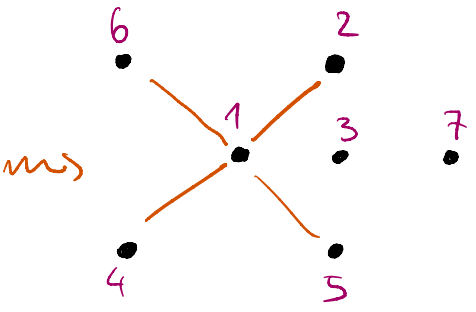
Example. *Find the tree associated to sequence* $P = (1,1,1,1,3) \in \{1,2,3,4,5,6,7\}^5$.





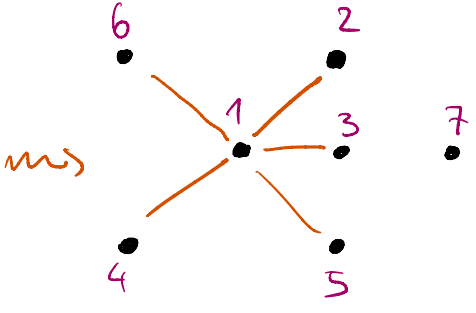
$P = (\cancel{1}, \cancel{1}, 1, 1, 3)$

$S = \{2, 4, 5\}$



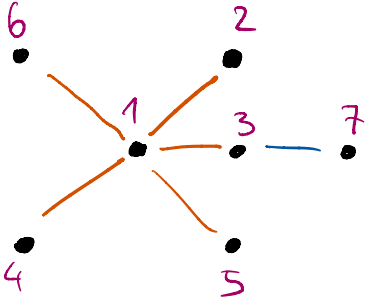
$P = (\cancel{1}, \cancel{1}, 1, 1, 3)$

$S = \{2, 4, 5, 6\}$

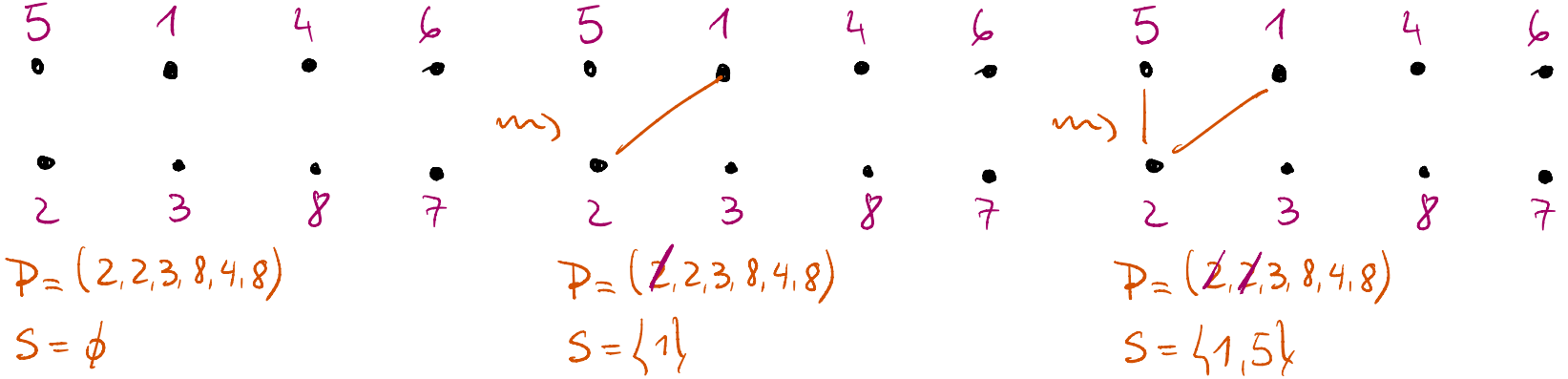


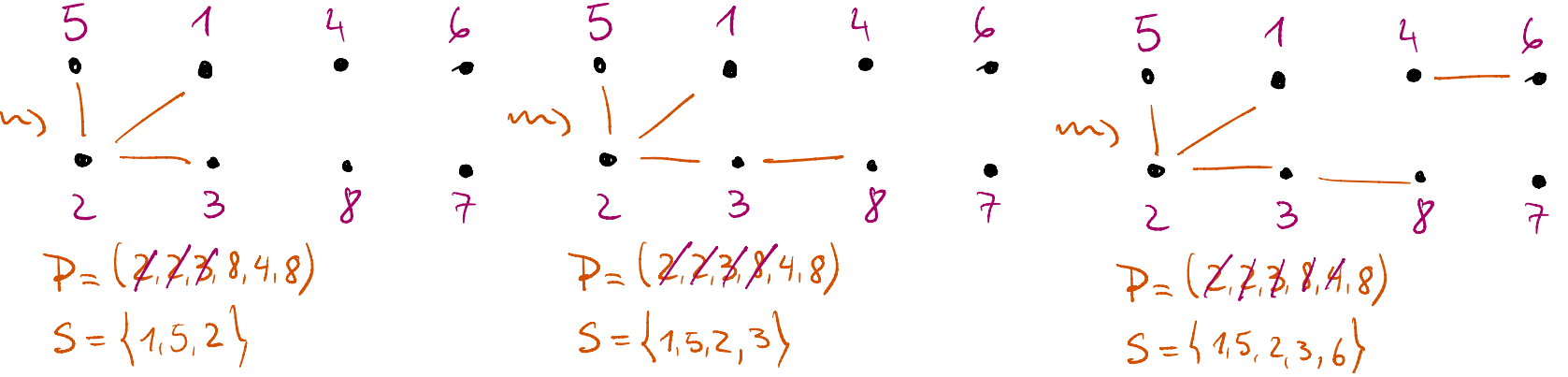
$P = (\cancel{1}, \cancel{1}, 1, \cancel{1}, \cancel{6})$

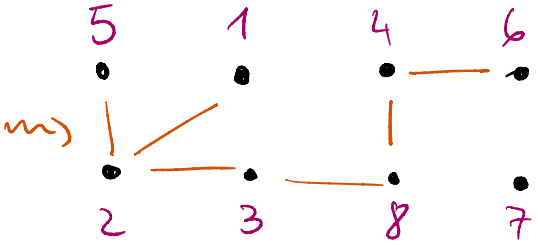
$S = \{2, 4, 5, 6, 1\} = V \setminus \{3, 7\}$



Example. *Find the tree associated to sequence* $\mathrm{P} = (2,2,3,8,4,8) \in \{1,2,3,4,5,6,7,8\}^6$.







$P = (2, 7, 3, 8, 4, 8)$

$S = \{1, 5, 2, 3, 6, 4\} = V \setminus \{7, 8\}$

